Image Enhancement in the Frequency Domain

Subtopic 3.5: Other Image Transforms

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Outline - Subtopic 3.5: Other Image Transforms

1. Introduction to Alternative Image Transforms
2. Hadamard Transform
3. Haar Transform
4. Discrete Cosine Transform (DCT)
5. Comparison of Transforms
6. Summary
7. References

Alternative Image Transforms: Overview

- **Why Other Transforms?**
  - Fourier transform not always optimal for all applications
  - Different transforms reveal different image properties
  - Some transforms more computationally efficient
  - Better suited for specific feature types
- **Key Alternatives:**
  1. Hadamard Transform
  2. Haar Transform
  3. Discrete Cosine Transform (DCT)
- **Common Characteristics:**
  - Orthogonal transformations
  - Energy compact representation
  - Real-valued coefficients
  - Efficient implementations possible

Comparison with Fourier Transform

- **Fourier Transform:**
  - Complex-valued coefficients
  - Global frequency information
  - Excellent for frequency analysis
  - No time/space localization
- **Alternative Transforms:**
  - Real-valued coefficients
  - More compact representation for certain signals
  - Better localization properties
  - More suitable for compression
  - Faster computation
- **Transform Properties to Consider:**
  - Separability
  - Orthogonality
  - Energy compaction
  - Computational complexity
  - Basis functions

Hadamard Transform: Overview

- **Definition:** Orthogonal transformation based on rectangular waveforms
- **Key Characteristics:**
  - Uses **Walsh functions** as basis
  - Only values: $+1$ and -1 (rectangular waveforms)
  - Real-valued and orthogonal
  - No multiplication needed (only addition/subtraction)
- **Advantages:**
  - Very fast computation
  - Simple implementation
  - No floating-point arithmetic
  - Good energy compaction
- **Applications:**
  - Image compression
  - Pattern recognition
  - Feature extraction
  - Real-time processing

Hadamard Transform: Mathematical Definition

- **1-D Forward Transform:**

$$F(u) = \frac{1}{N} \sum_{x=0}^{N-1} f(x) H_N(x, u) \tag{1}$$

where H_N is the Hadamard matrix of size $N \times N$

- **1-D Inverse Transform:**

$$f(x) = \sum_{u=0}^{N-1} F(u) H_N(x, u) \tag{2}$$

- **2-D Forward Transform:**

$$F(u, v) = \frac{1}{N^2} \sum_{x=0}^{N-1} \sum_{y=0}^{N-1} f(x, y) H_N(x, u) H_N(y, v) \tag{3}$$

- **Separable:** 2-D transform can be computed from 1-D transforms

Hadamard Matrix Construction

- **Recursive Definition:**

$$H_1 = \begin{bmatrix} 1 \end{bmatrix} \tag{4}$$

$$H_{2n} = \begin{bmatrix} H_n & H_n \\ H_n & -H_n \end{bmatrix} \tag{5}$$

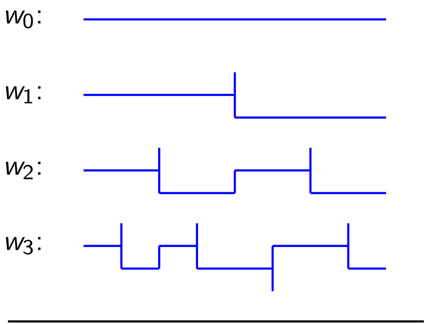
- **Example:** H_2 **and** H_4

$$H_2 = \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}, \quad H_4 = \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & -1 & 1 & -1 \\ 1 & 1 & -1 & -1 \\ 1 & -1 & -1 & 1 \end{bmatrix} \tag{6}$$

- **Key Properties:**
  - Orthogonal: $H \cdot H^T = N \cdot I$
  - Size must be power of 2: $N = 2^k$
  - Symmetric: $H = H^T$

Hadamard Transform: Basis Functions

Walsh Functions (Basis of Hadamard Transform)



Square Wave Basis Functions

Hadamard Transform: Computational Advantages

- **Complexity Comparison:**
  - FFT: $O(N \log N)$ with complex multiplications
  - Hadamard: $O(N \log N)$ with additions/subtractions only
  - Hadamard is **2-3 times faster** in practice
- **Implementation Advantages:**
  - Integer arithmetic only (no floating point)
  - Simple recursive structure
  - Suitable for hardware implementation
  - Low memory requirements
- **Limitations:**
  - Less intuitive than frequency domain
  - Image size must be power of 2
  - Less energy compaction than DCT
  - Not as widely used as FFT or DCT

Hadamard Transform: Applications

- **Image Compression:**
  - Energy is compact in low-frequency coefficients
  - Can discard high-frequency components
  - Simple to implement
  - Good compression ratios
- **Feature Extraction:**
  - Extract textural features
  - Pattern recognition
  - Dimensionality reduction
- **Image Enhancement:**
  - Contrast enhancement
  - Filtering in transform domain
  - Noise reduction
- **Real-time Applications:**
  - Video processing
  - Embedded systems
  - Fast processing requirement

Haar Transform: Overview

- **Definition:** Orthogonal transformation based on Haar wavelets
- **Key Characteristics:**
  - Uses **Haar wavelets** as basis functions
  - Similar to Hadamard but with hierarchical structure
  - Real-valued and orthogonal
  - Simple and elegant basis
- **Advantages:**
  - Excellent energy compaction
  - Good for analysis of discontinuities
  - Detects edges and sudden changes
  - Computationally efficient
  - Foundation of modern wavelets
- **Applications:**
  - Image compression (basis for JPEG2000)
  - Edge detection
  - Texture analysis
  - Hierarchical image representation

Haar Transform: Mathematical Definition

- **1-D Forward Transform:**

$$F(u) = \sum_{x=0}^{N-1} f(x) h_u(x) \tag{7}$$

where $h_u(x)$ is the u-th Haar wavelet function

- **Haar Wavelet Functions:**

$$h_u(x) = \begin{cases} \frac{1}{\sqrt{N}} & \text{for } x \in [0, N) \\ \frac{1}{\sqrt{N/2}} & \text{for approximation scale} \\ \pm\frac{1}{\sqrt{N/2}} & \text{for detail scale} \end{cases} \tag{8}$$

- **2-D Forward Transform:**

$$F(u, v) = \sum_{x=0}^{M-1} \sum_{y=0}^{N-1} f(x, y) h_u(x) h_v(y) \tag{9}$$

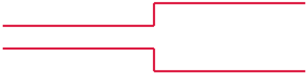
- **Separable:** 2-D transform from sequential 1-D transforms

Haar Basis Functions

Haar Wavelets and Scaling Functions

Scaling: 

$\phi(x)$ - Averaging

Mother: 

$\psi(x)$ - Detail extraction

Dilated: 

$\psi(2x)$, $\psi(2x-1)$ - Multiple scales

Haar wavelets provide local, hierarchical basis

Haar Transform: Multi-Scale Analysis

- **Hierarchical Decomposition:**
  - Approximation coefficients (low frequencies)
  - Detail coefficients (high frequencies)
  - Multiple scales simultaneously

- **Decomposition Process:**
  1. Approximate signal with half resolution
  2. Extract details as difference coefficients
  3. Recursively decompose approximation

- **Example: Decomposition Levels**
  - Level 1: Approximate + Detail at scale 1
  - Level 2: Approximate + Detail at scale 2
  - Level 3: Approximate + Detail at scale 3
  - ... and so on

Haar Transform: Advantages and Properties

- **Advantages:**
  - **Energy Compaction:** Excellent for natural images
  - **Localization:** Detects edges and discontinuities
  - **Hierarchical:** Multi-resolution representation
  - **Fast:** Similar to Hadamard complexity
  - **Simple:** Easy to understand and implement
- **Limitations:**
  - Basis functions have limited smoothness
  - Not ideal for smooth signals
  - Artifacts at discontinuities
  - Size must be power of 2
- **Comparison with Hadamard:**
  - Better energy compaction than Hadamard
  - Better edge detection than Hadamard
  - Foundation for more general wavelets

Haar Transform: Applications

- **Image Compression:**
  - Excellent energy compaction
  - Basis for JPEG2000 standard
  - High compression ratios
  - Progressive transmission
- **Edge and Feature Detection:**
  - Detail coefficients reveal edges
  - Discontinuity analysis
  - Multi-scale feature extraction
- **Image Denoising:**
  - Threshold detail coefficients
  - Preserves edges while removing noise
  - Multi-resolution denoising
- **Image Analysis:**
  - Texture analysis and classification
  - Fractal dimension estimation
  - Hierarchical feature extraction

Discrete Cosine Transform: Overview

- **Definition:** Orthogonal transformation using cosine basis functions
- **Key Characteristics:**
  - Uses cosine functions at different frequencies as basis
  - Real-valued and orthogonal
  - Excellent energy compaction
  - Boundary-friendly (even extension)
- **Advantages:**
  - **Best energy compaction** among all three transforms
  - Minimal boundary artifacts
  - Smooth basis functions
  - Most widely used in practice
- **Applications:**
  - JPEG image compression (most important)
  - Audio compression (MP3)
  - Video compression (MPEG, H.264)
  - General signal processing

Discrete Cosine Transform: Mathematical Definition

- **1-D Forward DCT:**

$$F(u) = c(u) \sum_{x=0}^{N-1} f(x) \cos\left(\frac{\pi(2x+1)u}{2N}\right) \tag{10}$$

where $c(u) = \sqrt{\frac{1}{N}}$ for $u = 0$ and $c(u) = \sqrt{\frac{2}{N}}$ for $u > 0$

- **1-D Inverse DCT:**

$$f(x) = \sum_{u=0}^{N-1} c(u) F(u) \cos\left(\frac{\pi(2x+1)u}{2N}\right) \tag{11}$$

- **2-D Forward DCT:**

$$F(u,v) = c(u)c(v) \sum_{x=0}^{M-1} \sum_{y=0}^{N-1} f(x,y) \cos\left(\frac{\pi(2x+1)u}{2M}\right) \cos\left(\frac{\pi(2y+1)v}{2N}\right) \tag{12}$$

DCT: Basis Functions

2-D DCT Basis Functions (8x8 Example)

Low Freq High Freq

B_{00}	B_{10}	B_{20}	B_{30}
B_{01}	B_{11}	B_{21}	B_{31}
B_{02}	B_{12}	B_{22}	B_{32}
B_{03}	B_{13}	B_{23}	B_{33}

High Freq

DCT basis functions show increasing spatial frequency

DCT: Key Properties

- **Orthogonality:**

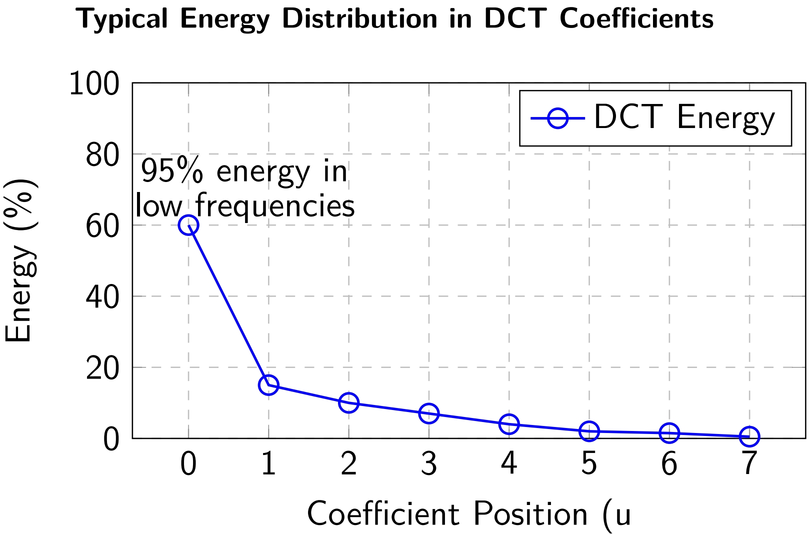
$$\sum_{x=0}^{N-1} \phi_u(x)\phi_v(x) = \delta_{uv} \tag{13}$$

where $\phi_u(x)$ is the u-th basis function and δ_{uv} is Kronecker delta

- **Separability:**
  - 2-D DCT can be computed as successive 1-D DCTs
  - Reduces complexity significantly
- **Energy Compaction:**
  - Most energy concentrated in low-frequency coefficients
  - Typically 95%+ energy in top-left coefficients
  - Best among orthogonal transforms
- **DC Component:**

$$F(0,0) = \frac{1}{M \cdot N} \sum_{x,y} f(x,y) \quad \text{(Average value)} \tag{14}$$

DCT: Energy Distribution



DCT provides excellent energy compaction compared to DFT

DCT: Advantages and Properties

- **Advantages:**
  - **Best Energy Compaction:** Superior to DFT, Hadamard, Haar
  - **No Boundary Effects:** Even extension minimizes artifacts
  - **Real-valued:** Simpler computation than complex FFT
  - **Smooth Basis:** Cosine functions are smooth and continuous
  - **Fast Algorithms:** Fast DCT similar to FFT complexity

- **Comparison with Other Transforms:**
  - Better than FFT for image compression
  - Better than Hadamard for energy compaction
  - Better than Haar for smooth signals
  - More computational cost than Hadamard

DCT: JPEG Compression Standard

- **Role in JPEG:**
  - Transform image into 8×8 blocks
  - Apply DCT to each block
  - Quantize high-frequency coefficients
  - Entropy coding (Huffman/Arithmetic)
  - Results in significant compression

- **Why DCT for JPEG?**
  1. Excellent energy compaction
  2. Standard block size: 8×8 optimal
  3. Fast algorithms available
  4. Minimal visual artifacts
  5. Natural boundary handling

- **Typical Compression Ratios:**
  - Quality 90: 10:1 compression
  - Quality 80: 20:1 compression
  - Quality 50: 50:1 compression

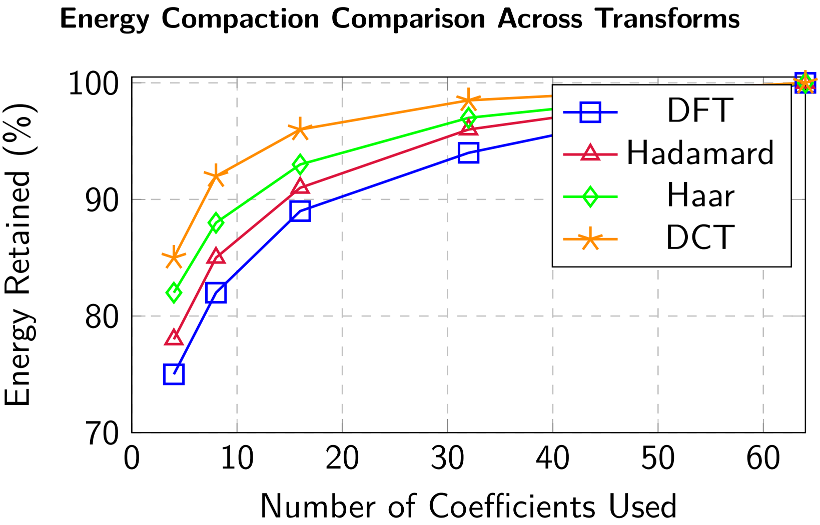
DCT: Applications

- **Image Compression:**
  - JPEG (most important application)
  - JPEG2000 (uses wavelets but DCT alternative)
  - Excellent lossy compression
- **Audio/Video Compression:**
  - MP3 audio compression
  - MPEG video standards
  - H.264/H.265 video compression
- **Image Enhancement:**
  - Frequency domain filtering
  - Noise reduction
  - Contrast enhancement
- **Feature Extraction:**
  - Texture analysis
  - Image classification
  - Content-based retrieval

Transform Comparison: Overview

Property	DFT	Hadamard	Haar	DCT
Basis	Exponential	Walsh	Wavelets	Cosine
Coefficients	Complex	Real	Real	Real
Orthogonal	Yes	Yes	Yes	Yes
Separable	Yes	Yes	Yes	Yes
Energy Compact	Moderate	Good	Very Good	Excellent
Edge Detection	Poor	Poor	Excellent	Moderate
Size Constraint	Any	Power of 2	Power of 2	Any
Computation	$O(N \log N)$	$O(N \log N)$	$O(N)$	$O(N \log N)$
Hardware	Complex	Simple	Simple	Moderate
Applications	Filtering	Real-time	Compression	Compression

Energy Compaction Comparison



DCT provides best energy compaction; DFT worst

Computational Complexity Comparison

- **Asymptotic Complexity:**

Transform	1-D	2-D
DFT/FFT	$O(N \log N)$	$O(N^2 \log N)$
Hadamard	$O(N \log N)$	$O(N^2 \log N)$
Haar	$O(N)$	$O(N^2)$
DCT	$O(N \log N)$	$O(N^2 \log N)$

- **Practical Speed (relative to DFT):**
  - DFT: $1\times$ (baseline)
  - Hadamard: 2-3$\times$ faster (simpler operations)
  - Haar: 5-10$\times$ faster (linear complexity)
  - DCT: Similar to DFT (but no complex arithmetic)

- **Fast Algorithms:**
  - DCT-II: Most common type
  - Similar acceleration as FFT
  - Can be computed from FFT

Selection Guide: When to Use Which Transform

Use DFT/FFT When:

- Frequency analysis needed
- Spectral properties important
- Filtering and convolution operations
- General signal processing

Use Haar When:

- Edge detection important
- Multi-scale analysis needed
- Progressive transmission required
- Image denoising

Use Hadamard When:

- Real-time processing required
- Hardware implementation preferred
- Integer arithmetic only
- Speed critical application

Use DCT When:

- Image compression required (MOST COMMON)
- Energy compaction critical
- Quality/speed balance needed
- Standard compression formats

Summary: Properties of Transforms

Property	**DFT**	**Hadamard**	**Haar**	**DCT**
Basis Type	Exponential	Rectangular	Wavelets	Cosine
Coefficient Type	Complex	Real	Real	Real
Orthogonality	Yes	Yes	Yes	Yes
Separability	Yes	Yes	Yes	Yes
Energy Compaction	Moderate	Good	Very Good	Excellent
Edge Detection	Poor	Poor	Excellent	Moderate
Boundary Effects	Significant	Moderate	Minimal	Minimal
Size Requirement	Any	Power of 2	Power of 2	Any
Computation	$O(N \log N)$	$O(N \log N)$	$O(N)$	$O(N \log N)$
Implementation	Complex	Simple	Simple	Moderate
Floating Point	Yes	No	No	Yes
Best Application	Filtering	Real-time	Wavelets	Compression

Implementation Considerations

- **DFT/FFT:**
  - Use standard FFT libraries (FFTW, LAPACK)
  - Requires complex arithmetic
  - Very optimized implementations available
- **Hadamard:**
  - Simple recursive implementation
  - Integer arithmetic only
  - Fast algorithm similar to FFT
- **Haar:**
  - Simple and efficient
  - Cascading filters (lifting scheme)
  - Foundation for wavelet transforms
- **DCT:**
  - Can use FFT-based computation
  - Specialized fast DCT algorithms
  - JPEG libraries use optimized DCT
  - Type-II DCT most common

Textbook and Reference Materials

Recommended Reading

- **Textbook:**
  - Rafael C. Gonzalez and Richard E. Woods, *Digital Image Processing*, Pearson Edition, Latest Edition.

- **Reference Books:**
  - I. Pitas, *Digital Image Processing Algorithms*, Prentice Hall, Latest Edition.
  - A. K. Jain, *Fundamentals of Digital Image Processing*, Prentice Hall of India Pvt. Ltd., Latest Edition.
  - K. Castleman, *Digital Image Processing*, Prentice Hall of India Pvt. Ltd., Latest Edition.

- **Additional Resources:**
  - Wavelet tutorials and references
  - JPEG compression standard documentation
  - Fast algorithm implementation guides

Key Takeaways - Alternative Image Transforms

- **Hadamard Transform:** Fast integer arithmetic; good for real-time processing
- **Haar Transform:** Excellent edge detection and multi-scale analysis
- **DCT Transform:** Best energy compaction; standard for image compression
- **Key Advantages Over DFT:**
  - Real-valued coefficients
  - Better energy compaction
  - Reduced computational requirements
  - Boundary-friendly properties

Selection Criteria:

Hadamard: Speed; Haar: Edges; DCT: Compression; FFT: Frequency analysis